

Brandon Y. Perez Martinez

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PROFESSIONAL SUMMARY

Self-taught AI and software engineer with 6 years of U.S. Air Force cyber operations experience and 4+ years building production agentic AI systems and quantitative engineering platforms. Builds multi-agent orchestration systems, custom MCP tooling, and fine-tuned LLM pipelines, alongside full-stack and cloud-native applications shipped to real users. DoD network operations background with two overseas deployments and an active CompTIA Security+.

TECHNICAL SKILLS

AI / Agentic Engineering: Multi-agent orchestration across mixed model fleets (Claude, Codex, ChatGPT, Grok, Antigravity), Model Context Protocol (MCP) server development, LLM fine-tuning (DPO/SFT) on open models (Qwen3-14B), local inference (Intel Arc), secure execution sandboxing (Kata Containers / microVMs for untrusted agent workloads), eval and trajectory analysis, prompt & tool-calling design

Languages: Python (advanced), TypeScript, Bash, SQL, PowerShell

Data & ML: PostgreSQL, TimescaleDB, LightGBM, scikit-learn, walk-forward validation, feature engineering, time-series pipelines, ETL, REST API integration

Full-Stack & Cloud: Next.js, React, Express, Drizzle ORM, Stripe, Cloudflare (Pages/Workers), Google Cloud Run, GCS, Docker, CI/CD (GitHub Actions, CodeQL, OpenSSF Scorecard)

Infra & DevOps: Linux (RHEL/Ubuntu/Debian), VPS fleet operations, systemd, cron automation, monitoring/failover, Git

Cybersecurity: Network security hardening, ACAS/Nessus, Active Directory, VMware/vSphere/ESXi, DoD network operations

EXPERIENCE

AI & Software Engineer | Independent / Self-Employed | Remote | Aug 2022 - Present

- Built a multi-agent AI orchestration system coordinating mixed fleets of Claude, Codex, ChatGPT, Grok, and Antigravity agents (planner + specialized sub-agents) for autonomous software engineering, research, and operations, using custom MCP servers exposing tools to the models, with fail-closed permission gating.
- Fine-tuned an open 14B model (Qwen3, LoRA SFT with a DPO/ORPO research branch) into an operator-gated security/pentest agent on local Intel Arc GPU hardware; built the trajectory-mining, preference-pair curation, and proof-safe evaluation-gate pipeline, including an automatic quarantine that blocks adapters emitting unsafe commands.
- Designed a secure execution layer running untrusted, agent-generated code no-egress inside Kata Containers microVMs, with a fail-closed admission gate: a production sandboxing pattern used by AI-infra companies.
- Designed and deployed a production algorithmic trading platform (Python, PostgreSQL/TimescaleDB): 80+ live data streams, 26+ concurrent strategies, regime-adaptive and Bayesian Kelly position sizing, a dedicated risk engine (circuit breakers, VaR/CVaR, correlation sizing), and genetic-algorithm strategy-weight evolution; execution across Hyperliquid, Kalshi, Polymarket, Coinbase, and Alpaca (live on Kalshi prediction markets, gated paper execution elsewhere). 1,100+ Python modules, 2,690+ commits, CI-gated.
- Built an ML signal pipeline (LightGBM ensemble, 742K+ candles, 45 features, walk-forward validation) with automated model lifecycle management.
- Operate full Linux VPS infrastructure: systemd services, 150+ cron jobs, real-time monitoring, automated failover and alerting.
- Built large-scale web data-extraction and enrichment pipelines (headless browser automation, reverse-engineered private/hidden APIs, Chrome DevTools Protocol) feeding a 100K+ record B2B database across 9 verticals with multi-signal scoring.

Full-Stack Engineer & Co-owner (15%) | Greenlightdate (greenlightdate.com) | 2026

- Built a production dating web application from scratch as co-owner and sole backend engineer: pnpm monorepo (Vite/React + Express + Drizzle/Postgres), Stripe billing, and OpenAPI → typed-client codegen.
- Executed a live cloud migration (Replit → Vercel + Cloud Run → Cloudflare Pages + GCS) with zero-downtime cutover.
- Hardened CI with CodeQL, dependency review, and OpenSSF Scorecard; delivered the application through a security audit and compliance review.

Cyber Systems Operations Technician | U.S. Air Force, 52nd Combat Communications Squadron | Aug 2016 - Jul 2022

- Engineered and fielded \$7.2M in Theater Deployable Communications (TDC) packages supporting global rapid-deployment missions.
- Administered and hardened Active Directory, VMware/vSphere/ESXi, and DoD network infrastructure; led ACAS/Nessus vulnerability scanning and remediation programs.
- **Syria (Operation Inherent Resolve, Oct 2018 - Feb 2019):** Restored 3,700+ mission-critical files after a system failure under deployed combat conditions, preserving continuity of operations.
- **Niger (Operation Juniper Shield, Aug 2020 - Mar 2021):** Delivered a \$9.6M network upgrade and eliminated 3,000+ vulnerabilities across 16 domains, hardening 104 virtualized systems.
- Senior Airman (SrA, E-4) at separation after six years of service.

SELECTED PROJECTS

Multi-Market Algorithmic Trading Platform: Production Python system trading crypto perps, prediction markets (Kalshi/Polymarket), and equities on one unified PostgreSQL/TimescaleDB pipeline. Risk engine (circuit breakers, VaR/CVaR, correlation sizing), genetic-algorithm strategy evolution, LightGBM signal ensemble; 1,100+ Python modules, 2,690+ commits, CI-gated, VPS-deployed with live execution on Kalshi and gated paper execution elsewhere.

Local LLM Pentest Agent (fine-tuning R&D): Fine-tuned Qwen3-14B (LoRA SFT plus a DPO/ORPO research branch) into an operator-gated Hack-The-Box agent on local Intel Arc GPU hardware. Built the full pipeline: trajectory mining, preference-pair curation, and a proof-safe evaluation gate that auto-quarantines unsafe adapters. Promoted adapter scored 21/22 on a held-out shadow eval. The completion ledger records 3 proof-safe confirmed end-to-end box completions.

Greenlightdate (Dating Web App, co-owner, live): Full-stack production app (greenlightdate.com), 15% equity, sole backend engineer: Vite/React + Express + Drizzle/Postgres, Stripe billing, OpenAPI → typed-client codegen. Executed a live Replit → Vercel+Cloud Run → Cloudflare Pages migration; CI hardened with CodeQL + OpenSSF Scorecard.

CERTIFICATIONS

- CompTIA Security+: ID# COMP001021119756 (renewed 2026, expires Feb 2029)